

Vakint

Single-scale vacuum-integral matching and evaluation

Version source: `crates/vakint/Cargo.toml` Owner: Vakint project

1 Overview

Vakint matches single-scale vacuum integrals to a library of known topologies, canonicalizes their momentum routing, reduces tensor numerators, and evaluates them analytically or numerically. Expressions are represented with Symbolica and contain a numerator multiplied by a `topo(...)` structure built from propagators.

Separate matching from evaluation

Canonical topology matching is useful on its own and is much cheaper than a complete evaluation. Tensor reduction and analytic backends invoke FORM; numerical sector decomposition invokes pySecDec and FORM. Choose the narrowest operation and evaluation order needed for the task.

1.1 Workflow

A complete calculation proceeds through explicit stages:

- create one Vakint engine and reuse it, because topology setup has a cost;
- parse or construct a Symbolica expression in Vakint's namespace;
- canonicalize the topology and momentum routing;
- tensor-reduce the numerator when required;
- evaluate the integral with the first supported backend in the configured order;
- substitute numerical parameters and external momenta when a numerical result is required.

`VakintExpression` is the structured sum of numerator/topology terms. A recognized topology can be rendered in canonical long or short form. If unknown integrals are allowed, matching may preserve them, but evaluation still needs a backend that supports the resulting topology.

1.2 External tools and reproducibility

Construction validates the selected backends

The default evaluation order includes AlphaLoop, MATAD, FMFT, and pySecDec. Settings validation therefore probes FORM and pySecDec even if a later example only canonicalizes an expression. FORM 4.2.1 or newer and pySecDec 1.6.4 or newer are the minimum versions checked by the current source. For a pure matching test, use an empty evaluation order in Rust; in Python, provide an empty `evaluation_order` when constructing Vakint.

Analytic AlphaLoop, MATAD, and FMFT methods depend on FORM. The pySecDec method depends on both FORM and pySecDec and requires numerical masses and, where applicable, external momenta. Record tool versions, normalization convention, epsilon depth, decimal precision, backend order, and temporary-output reuse policy with any result intended to be reproduced.

1.3 Test tiers and platform notes

The ordinary fast suite excludes pySecDec work while retaining verbose diagnostics and generated temporary files for inspection:

```
VAKINT_SKIP_PYSECDEC_TESTS=true RUST_BACKTRACE=full VAKINT_NO_CLEAN_TMP_DIR=T \  
RUST_LOG=DEBUG cargo test --package vakint --no-default-features
```

Pure parsing, matching, canonicalization, and settings tests belong in the normal CI tier. FORM-backed examples require a provisioned FORM job. pySecDec, MATAD, and FMFT comparisons are opt-in or scheduled because they depend on external installations and can be slow. On macOS, a GNU-compiler link failure mentioning missing `__emul...` symbols is handled by building with `EXTRA_MACOS_LIBS_FOR_GNU_GCC=T`, as consumed by Vakint's build script.

Vakint is used by GammaLoop for vacuum-integral work, but it owns its topology and evaluation conventions independently. Start with the [Vakint README](#) and generated catalogs.

2 Tutorial

This tutorial parses and canonicalizes one vacuum integral in Rust. It deliberately disables all evaluation backends, so the first success exercises Vakint's topology library and momentum routing without requiring FORM, MATAD, FMFT, or pySecDec.

2.1 Prerequisites

Use Rust 1.85 or newer. Clone the workspace, create a sibling binary project, and add the current Vakint crate by path:

```
git clone https://github.com/alphal00p/gammloop.git  
cargo new vakint-first-match  
cd vakint-first-match  
cargo add vakint --path ../gammloop/crates/vakint
```

Vakint uses Symbolica; ensure your use satisfies its license terms. External integration tools are unnecessary for the matching-only settings below. For a reproducible checkout, use the source revision shown in this page's footer.

Crates.io 0.1.2 uses the previous settings API

The published 0.1.2 crate stores settings in `Vakint::new(Some(settings))` and calls `to_canonical(input.as_view(), true)`. This manual follows the current workspace API, where settings are passed explicitly to each operation; the versioned reference is authoritative for whichever source revision you install.

2.2 Match a one-loop topology

Replace `src/main.rs` with:

```
use vakint::{  
    vakint_parse, EvaluationOrder, Vakint, VakintExpression, VakintSettings,  
};  
  
fn main() -> Result<(), Box<dyn std::error::Error>> {  
    let settings = VakintSettings {  
        allow_unknown_integrals: false,  
        evaluation_order: EvaluationOrder::empty(),  
        ..VakintSettings::default()  
    };  
    let vakint = Vakint::new()?;  
    vakint.validate_settings(&settings)?;  
  
    let input = vakint_parse!(
```

```

    "topo(prop(1,edge(1,1),k(1),muvsq,1))"
  )?;
  let canonical = vakint.to_canonical(
    &settings,
    input.as_view(),
    true,
  )?;

  println!("input: {}", VakintExpression::try_from(input)?);
  println!("canonical: {}", VakintExpression::try_from(canonical)?);
  Ok(())
}

```

Run `cargo run`. Success means Vakint recognizes the propagator as a supported one-loop vacuum topology, prints a canonical short form, and exits without probing an external executable. The canonical spelling may reorder identifiers or momentum routing; that normalized result, not the input's labels, is the comparison boundary.

An empty evaluation order is intentional

`EvaluationOrder::empty()` makes this a topology-matching tutorial. Default settings include analytic and numerical backends, and validating those settings checks their external dependencies even if you only planned to call `to_canonical`.

2.3 Move from matching to evaluation

Once the canonical form is understood, choose one supported backend explicitly, validate its dependencies, then apply the stages in order: tensor reduction, integral evaluation, and numerical substitution. Record the normalization factor, epsilon depth, decimal precision, backend settings, and tool versions with the result. Reuse one Vakint engine because topology construction has a cost.

2.4 Troubleshooting and next steps

- An unrecognized-integral error means the propagators, powers, masses, or momentum routing did not match a registered topology and `allow_unknown_integrals` is false. Print the long form and compare it with the generated topology table.
- A FORM or pySecDec version error means a nonempty evaluation order slipped into the settings; return to `EvaluationOrder::empty()` for pure canonicalization.
- If two inputs look different after matching, compare their canonical forms rather than their original propagator numbering.
- Continue with the topology-and-evaluation manual before enabling a backend or adding tensor numerators.

Ground truth for this walkthrough

The workflow follows Vakint's maintained Rust example and the matching path in `Vakint::to_canonical`.

3 Topology matching, reduction, and evaluation

Vakint recognizes an integral only after its propagators, masses, powers, and momentum routing can be mapped to a supported topology pattern. The generated topology table in this manual is constructed from the engine's registered topology library, so it stays aligned with the build instead of being maintained as a second handwritten list.

3.1 Parsing and topology matching

A `VakintExpression` is a sum of numerator/topology pairs. Parsing checks the Vakint namespace; matching then searches allowed substitutions and momentum shifts. Unknown-topology handling is a setting: preserving an unknown term is useful for partial simplification, but does not make that term evaluable.

Canonical long form is explicit and suitable for auditing. Canonical short form is convenient for display and library lookup. Both represent the same matched topology only after canonicalization has succeeded.

3.2 Normalization and canonicalization

Normalization conventions cover loop-measure factors, mass scales, epsilon powers, and the requested expansion depth. Set them before comparing results from different backends. Momentum canonicalization may rename loop variables and reorder propagators; compare canonical expressions rather than input spelling when testing equivalence.

Canonicalization is a pure CI boundary

Parsing, matching, normalization, and canonical routing can be tested without FORM or pySecDec when the evaluation order is empty. Keep these examples in the ordinary test tier so topology regressions do not depend on external executables.

3.3 Tensor reduction

Tensor numerators are reduced to scalar integrals before analytic evaluation where required. Reduction depends on the topology, Lorentz rank, dimension convention, and scalar-product normalization. FORM-backed reduction belongs to the provisioned integration tier; retain the generated FORM input and temporary directory when diagnosing a mismatch.

3.4 Evaluation order and backends

The evaluation order is an ordered fallback policy, not a request to combine backend results. Vakint tries each configured method whose capability matches the canonical topology and stops at the first successful evaluation. AlphaLoop, MATAD, and FMFT are analytic FORM-backed paths. pySecDec is numerical and additionally needs numeric parameters and external momenta where the topology requires them.

For reproducible comparisons record:

- the canonical topology and normalization convention;
- epsilon expansion depth and decimal precision;
- the ordered backend list and backend-specific settings;
- FORM, pySecDec, MATAD, and FMFT versions where applicable;
- parameter substitutions and any retained temporary-output directory.

Python is an embedded community module

`symbolica.community.vakint` is registered into a Symbolica installation; it is not a standalone PyPI package. Constructing its `Vakint` class validates the configured backends. Pass an empty evaluation order for pure matching work on machines without FORM/pySecDec.

The supported topology definitions are registered by Vakint's topology source, and minimum external versions are read directly into the generated dependency table.

4 Rust and Python API boundaries

Generated-reference contract

Signatures, fields, variants, defaults, feature gates, and source locations come from ordered generated

descriptors. Hand-written prose explains intent and workflow; it must not duplicate a signature as a second source of truth. Rust documentation explicitly marked as Typst markup may be evaluated as markup. Python docstrings and all other external metadata are escaped as plain text.

Rust scope: vakint

Python scope: symbolica.community.vakint

4.1 Rust package

The main Rust types make the workflow state explicit:

- Vakint owns the generated topology library and matching/evaluation operations;
- VakintSettings owns symbols, precision, normalization, tool paths, validation, and backend order;
- VakintExpression and VakintTerm separate numerators from topology atoms;
- EvaluationOrder and EvaluationMethod select AlphaLoop, MATAD, FMFT, or pySecDec;
- NumericalEvaluationResult stores Laurent coefficients in the dimensional regulator.

For a dependency-free canonicalization setup, select an empty backend order before validating:

```
use vakint::{EvaluationOrder, Vakint, VakintSettings};

let engine = Vakint::new()?;
let settings = VakintSettings {
    evaluation_order: EvaluationOrder::empty(),
    ..VakintSettings::default()
};
engine.validate_settings(&settings)?;
```

Parsing macros and helpers build Vakint-namespaced Symbolica atoms. The generated catalog owns their exact grammar and error types. Do not interpolate untrusted input into a macro example; use the runtime parser path when the expression arrives dynamically.

4.2 Python community module

Community module and feature pairing

Python users import symbolica.community.vakint. It is not a standalone Vakint wheel. The Rust build must include symbolica_community_module; python_stubgen generates type metadata but does not, by itself, enable the community module in the current feature graph.

Install a Symbolica Python distribution only when its release includes Vakint and verify it with `python -c "import symbolica.community.vakint";` Vakint is not installed by a separate wheel in this repository. Source embedders add the crate to the external symbolica-community assembly, enable symbolica_community_module, and invoke VakintWrapper's SymbolicaCommunityModule registration while building the Symbolica extension. Stub generation describes that mounted surface but never mounts it into an existing Python installation.

The module exports four classes: Vakint, VakintExpression, VakintEvaluationMethod, and VakintNumericalResult. This example constructs a matching-only engine without probing external backends:

```
from symbolica.community.vakint import Vakint

vakint = Vakint(evaluation_order=[])
# Supply a Symbolica Expression in the Vakint namespace to `to_canonical`.
canonical = vakint.to_canonical(expr, short_form=True)
```

Vakint exposes canonicalization, tensor reduction, integral evaluation, complete evaluation, and numerical conversion. `VakintEvaluationMethod` constructs configured backend choices. `VakintNumericalResult` converts Laurent coefficients to Python tuples and compares results with thresholds and optional errors.

Construct one engine and reuse it. When enabling `pySecDec`, give explicit numerical masses and external momenta and decide whether generated output may be reused. Do not run those heavy, external-tool examples as part of an ordinary documentation render.

Source starting points are [the Rust API](#) and [the Python module](#).

5 Releases and change history

Release-history policy

Vakint does not currently have a canonical `CHANGELOG.md` in this repository. The selected release is identified by `crates/vakint/Cargo.toml`, repository tags, and the exact source revision. This manual must not synthesize historical entries from commits or GammaLoop's release history.

A future curated changelog should distinguish topology-library changes, canonicalization and tensor-reduction changes, analytic backend changes, `pySecDec` integration, normalization and precision changes, and Rust/Python API migrations. These categories can affect reproducibility even when a high-level method name is unchanged.

Documentation may publish immutable snapshots for Vakint tags and a moving `latest` channel. Every snapshot records the crate version, feature set, FORM and `pySecDec` prerequisites, and repository revision. Until curated release notes exist, verify behavior against the selected source and tests beginning at [the maintained README](#).

Supported API catalog

This curated catalog is generated from explicit ordered scopes. Links lead to the exhaustive language references.

vakint

Vakint’s supported topology and evaluation API. Generated topology and backend tables are *validated* against runtime sources.

Vakint expression parser

Parses a Symbolica expression using Vakint's namespace by default or a caller-supplied namespace.

Parses a Symbolica expression using Vakint's namespace by default or a caller-supplied namespace.

```
#[macro_export] macro_rules! vakint_parse
{
    ($s: expr) =>
    {{ symbolica::try_parse!($s, default_namespace = $crate::NAMESPACE) }};
    ($s: expr, $ns: expr) =>
    {{ symbolica::try_parse!($s, default_namespace = $ns) }};
}
```

Supported usage

```
let expression = vakint::vakint_parse!("x").expect("valid Vakint expression"); let
explicit = vakint::vakint_parse!("x", vakint::NAMESPACE).expect("valid explicit
namespace");
```

Exhaustive reference · Source

Vakint symbol parser

Resolves a symbol name in Vakint's namespace while preserving an explicit namespace.

Resolves a symbol name in Vakint's namespace while preserving an explicit namespace.

```
#[macro_export] macro_rules! vakint_symbol
{
    ($s: expr) =>
    {{
        if
        format!("{}",
        $s).starts_with(format!("{}", $crate::NAMESPACE).as_str())
        {
            match symbolica::try_parse!($s)
            {
                Ok:::<_, String>(a) => match a
                {
                    symbolica::atom::Atom::Var(sym) => sym.get_symbol(), _ =>
                    panic!("Parsed atom is not a symbol: {}. ", $s),
                }, Err(e) =>
                panic!("Could not parse symbol from string: {}, Error: {}",
                $s, e),
            }
        } else
        {
            match
            symbolica::try_parse!(format!("{}", $s), default_namespace =
            $crate::NAMESPACE)
```

}

Supported usage

```
let symbol = vakint::vakint_symbol!("mass");
```

Exhaustive reference · Source

Vakint engine

Owns supported topology matching, canonicalization, reduction, and evaluation operations.

Owns supported topology matching, canonicalization, reduction, and evaluation operations.

```
#[derive(Debug, Clone)] pub struct Vakint { pub topologies : Topologies, }
```

Supported usage

```
let engine = vakint::Vakint::new()?;
```

topologies

Kind: Field

Topologies

Exhaustive reference · Source

Vakint settings

Controls symbols, precision, normalization, external tools, and backend evaluation order.

Controls symbols, precision, normalization, external tools, and backend evaluation order.

```
#[derive(Debug, Clone)] pub struct VakintSettings
{
    #[allow(unused)] pub epsilon_symbol : String, pub mu_r_sq_symbol : String,
    pub form_exe_path : String, pub python_exe_path : String, pub
    verify_numerator_identification : bool, pub integral_normalization_factor
    : LoopNormalizationFactor, pub run_time_decimal_precision : u32, pub
    allow_unknown_integrals : bool, pub clean_tmp_dir : bool, pub
    evaluation_order : EvaluationOrder, pub
    number_of_terms_in_epsilon_expansion : i64, pub
    precision_for_input_float_rationalization :
    InputFloatRationalizationPrecision, pub use_dot_product_notation : bool,
    pub temporary_directory : Option < String > ,
}
```

Supported usage


```
let settings = vakint::VakintSettings::default();
```

epsilon_symbol

Kind: Field

String

mu_r_sq_symbol

Kind: Field

String

form_exe_path

Kind: Field

String

python_exe_path

Kind: Field

String

verify_numerator_identification

Kind: Field

bool

integral_normalization_factor

Kind: Field

LoopNormalizationFactor

run_time_decimal_precision

Kind: Field

u32

allow_unknown_integrals

Kind: Field

bool

clean_tmp_dir

Kind: Field

bool

evaluation_order

Kind: Field

EvaluationOrder

number_of_terms_in_epsilon_expansion

Kind: Field

i64

precision_for_input_float_rationalization

Kind: Field

InputFloatRationalizationPrecision

use_dot_product_notation

Kind: Field

bool

temporary_directory

Kind: Field

Option < String >

Exhaustive reference · Source

Vakint expression

Stores a sum of matched vacuum-integral terms with separate numerators.

Stores a sum of matched vacuum-integral terms with separate numerators.

```
#[derive(Debug, Clone, PartialEq, Eq)] pub struct
VakintExpression(pub Vec < VakintTerm >);
```

Supported usage

```
use vakint::VakintExpression;
```

0

Kind: Field

Vec < VakintTerm >

Exhaustive reference · Source

Evaluation order

Defines the ordered analytic and numerical backends available to an evaluation.

Defines the ordered analytic and numerical backends available to an evaluation.

```
#[derive(Debug, Clone)] pub struct
EvaluationOrder(pub Vec < EvaluationMethod >);
```

Supported usage

```
let order = vakint::EvaluationOrder::empty();
```

0

Kind: Field

Vec < EvaluationMethod >

Exhaustive reference · Source

Canonicalize an expression

Matches and rewrites each integral into Vakint's canonical topology representation.

Matches and rewrites each integral into Vakint's canonical topology representation.

```
fn
canonicalize(& mut self, settings : & VakintSettings, topologies : &
Topologies, short_form : bool,) -> Result < (), VakintError >
```

Parameter	Type
self	& mut self

Parameter	Type
settings	& VakintSettings
topologies	& Topologies
short_form	bool

Returns: Result < (), VakintError >

Supported usage

```
use vakint::VakintExpression;
```

Exhaustive reference · Source

Tensor-reduce an expression

Reduces tensor numerators to scalar-integral combinations.

Reduces tensor numerators to scalar-integral combinations.

```
fn tensor_reduce(& mut self, vakint : & Vakint, settings : & VakintSettings,)
-> Result < (), VakintError >
```

Parameter	Type
self	& mut self
vakint	& Vakint
settings	& VakintSettings

Returns: Result < (), VakintError >

Supported usage

```
use vakint::VakintExpression;
```

Exhaustive reference · Source

Evaluate an expression

Evaluates canonical scalar integrals using the configured ordered backend policy.

Evaluates canonical scalar integrals using the configured ordered backend policy.

```
fn
evaluate_integral(& mut self, vakint : & Vakint, settings : & VakintSettings,)
-> Result < (), VakintError >
```

Parameter	Type
self	& mut self
vakint	& Vakint
settings	& VakintSettings

Returns: Result < (), VakintError >

Supported usage

```
use vakint::VakintExpression;
```

Exhaustive reference · Source

Numerical Laurent result

Stores numerical Laurent coefficients by power of the dimensional regulator.

Stores numerical Laurent coefficients by power of the dimensional regulator.

```
#[derive(Debug, Clone, Default)] pub struct  
NumericalEvaluationResult(pub Vec < (i64, Complex < Float >) >);
```

Supported usage

```
use vakint::NumericalEvaluationResult;
```

0

Kind: Field

Vec < (i64, Complex < Float >) >

Exhaustive reference · Source

vakint-community

Exports registered by vakint.

Vakint

Base class of vakint engine, from which all vakint functions are initiated.

Base class of vakint engine, from which all vakint functions are initiated.

It is best to create a single instance of this class and reuse it for multiple evaluations,

as the setup of the instance can be time consuming since it involves the processing of all known topologies.

```
class Vakint:
```

Requires features: symbolica_community_module, python_stubgen

Inspect the registered export

```
from symbolica.community.vakint import Vakint  
help(Vakint)
```

__new__

Kind: AssociatedFunction

```
def __new__(cls, run_time_decimal_precision: typing.Optional[builtins.int] = None,  
evaluation_order: typing.Optional[typing.Sequence[VakintEvaluationMethod]] = None,  
epsilon_symbol: typing.Optional[Expression] = None, mu_r_sq_symbol:  
typing.Optional[Expression] = None, form_exe_path: typing.Optional[builtins.str] =  
None, python_exe_path: typing.Optional[builtins.str] = None,  
verify_numerator_identification: typing.Optional[builtins.bool] = None,  
integral_normalization_factor: typing.Optional[builtins.str] = None,  
allow_unknown_integrals: typing.Optional[builtins.bool] = None, clean_tmp_dir:  
typing.Optional[builtins.bool] = None, number_of_terms_in_epsilon_expansion:  
typing.Optional[builtins.int] = None, use_dot_product_notation:  
typing.Optional[builtins.bool] = None, temporary_directory:  
typing.Optional[builtins.str] = None) -> Vakint:
```

Create a new Vakint instance, specifying details of the evaluation stack. Note that the same instance can be recycled across multiple evaluations.

Note that the creation of a Vakint instance involves the processing and creation of the library of all known topologies, which can be time consuming.

Examples

```
```python
```

```
vakint = Vakint(
integral_normalization_factor="MSbar",
```

```

mu_r_sq_symbol=S("mursq"),
If you select 5 terms, then MATAD will be used, but for 4 and fewer, alphaloop is
will be used as
it is first in the evaluation_order supplied.
number_of_terms_in_epsilon_expansion=4,
evaluation_order=[
VakintEvaluationMethod.new_alphaloop_method(),
VakintEvaluationMethod.new_matad_method(),
VakintEvaluationMethod.new_fmft_method(),
VakintEvaluationMethod.new_pysecdec_method(
min_n_evals=10_000,
max_n_evals=1000_000,
numerical_masses=masses,
numerical_external_momenta=external_momenta
),
],
form_exe_path="form",
python_exe_path="python3",
)
...

```

## Parameters

-----

```

run_time_decimal_precision : Optional[int]
The decimal precision to be used during the evaluation. Default is 17.
evaluation_order : Optional[Sequence[VakintEvaluationMethod]]
A list of `VakintEvaluationMethod` instances specifying the order in which evaluation
methods are to be applied. Default is all available methods in a sensible order.
epsilon_symbol : Optional[Expression]
The symbol to be used for the dimensional regularisation parameter epsilon. Default
is " ϵ ".
mu_r_sq_symbol : Optional[Expression]
The symbol to be used for the renormalisation scale squared. Default is "mursq".
form_exe_path : Optional[str]
The path to the FORM executable. Default is "form".
python_exe_path : Optional[str]
The path to the Python executable. Default is "python3".
verify_numerator_identification : Optional[bool]
Whether to verify the identification of numerator structures. Default is True.
integral_normalization_factor : Optional[str]
The normalization factor to be used for integrals. Can be "MSbar", "pySecDec",
"FMFTandMATAD" or a custom string. Default is "MSbar".
allow_unknown_integrals : Optional[bool]
Whether to allow unknown integrals to be processed. Default is True.
clean_tmp_dir : Optional[bool]
Whether to clean the temporary directory after evaluation. Default is True, unless
the environment variable VAKINT_NO_CLEAN_TMP_DIR is set.
number_of_terms_in_epsilon_expansion : Optional[int]
The number of terms in the epsilon expansion to be computed. Default is 4.
use_dot_product_notation : Optional[bool]
Whether to use dot product notation for scalar products. Default is False.
temporary_directory : Optional[str]
The path to the temporary directory to be used. Default is None, in which case a
system temporary directory will be used.

```

### **run\_time\_decimal\_precision**

**Kind:** Parameter

run\_time\_decimal\_precision: typing.Optional[builtins.int] = None

**Default:** None

The decimal precision to be used during the evaluation. Default is 17.

### **evaluation\_order**

**Kind:** Parameter

evaluation\_order: typing.Optional[typing.Sequence[VakintEvaluationMethod]] = None

**Default:** None

A list of `VakintEvaluationMethod` instances specifying the order in which evaluation methods are to be applied. Default is all available methods in a sensible order.

### **epsilon\_symbol**

**Kind:** Parameter

epsilon\_symbol: typing.Optional[Expression] = None

**Default:** None

The symbol to be used for the dimensional regularisation parameter epsilon. Default is " $\epsilon$ ".

### **mu\_r\_sq\_symbol**

**Kind:** Parameter

mu\_r\_sq\_symbol: typing.Optional[Expression] = None

**Default:** None

The symbol to be used for the renormalisation scale squared. Default is "mursq".

### **form\_exe\_path**

**Kind:** Parameter

form\_exe\_path: typing.Optional[builtins.str] = None

**Default:** None

The path to the FORM executable. Default is "form".

### **python\_exe\_path**

**Kind:** Parameter

python\_exe\_path: typing.Optional[builtins.str] = None

**Default:** None

The path to the Python executable. Default is "python3".

### **verify\_numerator\_identification**

**Kind:** Parameter

verify\_numerator\_identification: typing.Optional[builtins.bool] = None

**Default:** None

Whether to verify the identification of numerator structures. Default is True.

**integral\_normalization\_factor**

**Kind:** Parameter

integral\_normalization\_factor: typing.Optional[builtins.str] = None

**Default:** None

The normalization factor to be used for integrals. Can be "MSbar", "pySecDec", "FMFTandMATAD" or a custom string. Default is "MSbar".

**allow\_unknown\_integrals**

**Kind:** Parameter

allow\_unknown\_integrals: typing.Optional[builtins.bool] = None

**Default:** None

Whether to allow unknown integrals to be processed. Default is True.

**clean\_tmp\_dir**

**Kind:** Parameter

clean\_tmp\_dir: typing.Optional[builtins.bool] = None

**Default:** None

Whether to clean the temporary directory after evaluation. Default is True, unless the environment variable VAKINT\_NO\_CLEAN\_TMP\_DIR is set.

**number\_of\_terms\_in\_epsilon\_expansion**

**Kind:** Parameter

number\_of\_terms\_in\_epsilon\_expansion: typing.Optional[builtins.int] = None

**Default:** None

The number of terms in the epsilon expansion to be computed. Default is 4.

**use\_dot\_product\_notation**

**Kind:** Parameter

use\_dot\_product\_notation: typing.Optional[builtins.bool] = None

**Default:** None

Whether to use dot product notation for scalar products. Default is False.

**temporary\_directory**

**Kind:** Parameter

temporary\_directory: typing.Optional[builtins.str] = None

**Default:** None

The path to the temporary directory to be used. Default is None, in which case a system temporary directory will be used.

**numerical\_result\_from\_expression**

**Kind:** Method

```
def numerical_result_from_expression(self, expr: Expression) ->
VakintNumericalResult:
```

Convert a Symbolica expression to a vakint numerical result, interpreting the expression as a Laurent series in the dimensional regularisation parameter epsilon.

```
Examples
```python
res = vakint.numerical_result_from_expression(E("vakint::ε^-2 + 1 +
0.12*vakint::ε^-1"))
str(res)
# ε^-2 : (1.0000000000000000+0i)
# ε^-1 : (1.2000000000000000e-1+0i)
# ε^ 0 : (1.0000000000000000+0i)
```
```

Parameters

-----

**expr** : Expression

A Symbolica expression representing a Laurent series in the dimensional regularisation parameter epsilon specified in the vakint engine.

**expr**

**Kind:** Parameter

expr: Expression

A Symbolica expression representing a Laurent series in the dimensional regularisation parameter epsilon specified in the vakint engine.

**numerical\_evaluation**

**Kind:** Method

```
def numerical_evaluation(self, evaluated_integral: typing.Any, params:
typing.Mapping[builtins.str, builtins.float], externals:
typing.Optional[typing.Mapping[builtins.int, tuple[builtins.float, builtins.float,
builtins.float, builtins.float]]] = None) -> tuple[VakintNumericalResult,
typing.Optional[VakintNumericalResult]]:
```

Perform a numerical evaluation of an integral parameterically evaluated by Vakint, given numerical values for all parameters and optionally for external momenta.

## Examples

```
```python
evaluated_integral =
vakint.evaluate_integral(E("(k(1,11)*k(1,11)+p(1,12)*p(2,12))*topo(prop(1,edge(1,1),k(1),muvsq,1))"
default_namespace="vakint"))
numerical_result, numerical_error = vakint.numerical_evaluation(
evaluated_integral,
{ "muvsq": 2., "mursq": 3.},
{
1: (0.1, 0.2, 0.3, 0.4),
2: (0.5, 0.6, 0.7, 0.8)
}
)
str(numerical_result)
# ε^-1 : (0+27.63489232305020i)
# ε^ 0 : (0+-62.73919274007806i)
# ε^+1 : (0+107.7642844179578i)
# ε^+2 : (0+-138.7269737122023i)
```
```



```
...
```

## Parameters

-----

**evaluated\_integral** : Expression

A Symbolica expression representing an integral that has been evaluated parameterically by Vakint

**params** : Dict[str, float]

A dictionary mapping parameter names to their numerical values.

**externals** : Optional[Dict[int, Tuple[float, float, float, float]]]

An optional dictionary mapping external momentum indices to their numerical 4-vector values.

### **evaluated\_integral**

**Kind:** Parameter

evaluated\_integral: typing.Any

A Symbolica expression representing an integral that has been evaluated parameterically by Vakint

### **params**

**Kind:** Parameter

params: typing.Mapping[builtins.str, builtins.float]

A dictionary mapping parameter names to their numerical values.

### **externals**

**Kind:** Parameter

externals: typing.Optional[typing.Mapping[builtins.int, tuple[builtins.float, builtins.float, builtins.float, builtins.float]]] = None

**Default:** None

An optional dictionary mapping external momentum indices to their numerical 4-vector values.

### **numerical\_result\_to\_expression**

**Kind:** Method

def numerical\_result\_to\_expression(self, result: VakintNumericalResult) -> Expression:

Convert a vakint numerical result back to a Symbolica expression representing a Laurent series in the dimensional regularisation parameter epsilon.

## Examples

```python

evaluated_integral =

vakint.evaluate_integral(E("(k(1,11)*k(1,11)+p(1,12)*p(2,12))*topo(prop(1,edge(1,1),k(1),muvsq,1))", default_namespace="vakint"))

numerical_result, numerical_error = vakint.numerical_evaluation(
evaluated_integral,

{ "muvsq": 2., "mursq": 3.},

{

1: (0.1, 0.2, 0.3, 0.4),

2: (0.5, 0.6, 0.7, 0.8)

```

}
)
vakint.numerical_result_to_expression(numerical_result)
#
107.7642844179578i*ε+27.63489232305020i*ε^-1+-138.7269737122023i*ε^2+-62.73919274007806i

```

result

Kind: Parameter

result: VakintNumericalResult

to_canonical

Kind: Method

```

def to_canonical(self, integral_expression: Expression, short_form:
typing.Optional[builtins.bool] = None) -> Expression:

```

Convert a vakint expression to its canonical form, optionally using a short form for the topology representation.

Examples

```

```python
integral_expr =
E("(k(99,11)*k(99,11)+p(1,12)*p(2,12))*topo(prop(18,edge(7,7),k(99),muvsq,1))",
default_namespace="vakint")
vakint.to_canonical(integral_expr)
(k(1,11)^2+p(1,12)*p(2,12))*topo(prop(1,edge(1,1),k(1),muvsq,1))
vakint.to_canonical(integral_expr,short_form=True)
(k(1,11)^2+p(1,12)*p(2,12))*topo(I1L(muvsq,1))
```

```

Parameters

integral_expression : Expression

A Symbolica expression representing a vakint integral.

short_form : Optional[bool]

Whether to use the short form for the topology representation. Default is False.

integral_expression

Kind: Parameter

integral_expression: Expression

A Symbolica expression representing a vakint integral.

short_form

Kind: Parameter

short_form: typing.Optional[builtins.bool] = None

Default: None

Whether to use the short form for the topology representation. Default is False.

tensor_reduce

Kind: Method

```

def tensor_reduce(self, integral_expression: Expression) -> Expression:

```

Convert a vakint expression to a form where tensor integrals are reduced to scalar integrals.

```
## Examples
```python
integral_expr =
E("(k(1,11)*k(1,11)+p(1,12)*k(1,12)+k(1,101)*k(1,102))*topo(prop(1,edge(1,1),k(1),muvsq,1))",
default_namespace="vakint")

vakint.tensor_reduce(integral_expr)
(k(1,1)^2-(2*ε-4)^-1*k(1,1)^2*g(101,102))*topo(prop(1,edge(1,1),k(1),muvsq,1))
```
```

Parameters

integral_expression : Expression

A Symbolica expression representing a vakint integral.

integral_expression

Kind: Parameter

integral_expression: Expression

A Symbolica expression representing a vakint integral.

evaluate_integral

Kind: Method

def evaluate_integral(self, integral_expression: Expression) -> Expression:

Perform the parametric evaluation of *only the integral* appearing in the Symbolica expression given in input representing a vakint integral.

The numerator is left unchanged.

```
## Examples
```python
integral_expr =
E("(k(1,11)*k(1,11)+p(1,12)*k(1,12)+k(1,101)*k(1,102))*topo(prop(1,edge(1,1),k(1),muvsq,1))",
default_namespace="vakint")

vakint.evaluate_integral(integral_expr)
ε*(-((-π^2*i*log(π)+π^2*i*log(1/4*π^-1*mursq)))*(muvsq^2+muvsq*k(1,1)*p(1,1)+[...])
```
```

Parameters

integral_expression : Expression

A Symbolica expression representing a vakint integral.

integral_expression

Kind: Parameter

integral_expression: Expression

A Symbolica expression representing a vakint integral.

evaluate

Kind: Method

```
def evaluate(self, integral_expression: Expression) -> Expression:
```

Perform the complete parametric evaluation of the vaking integral represented by the Symbolica expression given in input.

Note that the tensor reduction will be automatically performed on the input given.

Examples

```
```python
```

```
integral_expr =
```

```
E("(k(1,11)*k(1,11)+p(1,12)*k(1,12)+k(1,101)*k(1,102))*topo(prop(1,edge(1,1),k(1),muvsq,1))",
default_namespace="vakint")
```

```
vakint.evaluate(integral_expr)
```

```
#
```

```
 $\epsilon*((\mu v s q^2 + 1/4 * \mu v s q^2 * g(101, 102)) * (1/2 * \pi^2 * i * \log(\pi)^2 + 1/2 * \pi^2 * i * \log(1/4 * \pi^2 - 1 * \mu r s q)^2 -$
[...])
```

```
```
```

Parameters

integral_expression : Expression

A Symbolica expression representing a vakint integral.

integral_expression

Kind: Parameter

integral_expression: Expression

A Symbolica expression representing a vakint integral.

Exhaustive reference · Source

VakintEvaluationMethod

Class representing a vakint evaluation method, which can be used to specify the evaluation order in a Vakint instance.

Class representing a vakint evaluation method, which can be used to specify the evaluation order in a Vakint instance.

```
class VakintEvaluationMethod:
```

Requires features: symbolica_community_module, python_stubgen

Inspect the registered export

```
from symbolica.community.vakint import VakintEvaluationMethod  
help(VakintEvaluationMethod)
```

__str__

Kind: Method

```
def __str__(self) -> builtins.str:
```

String representation of the evaluation method.

new_alphaloop_method

Kind: AssociatedFunction

```
def new_alphaloop_method(cls) -> VakintEvaluationMethod:
```

Create a new VakintEvaluationMethod instance representing the AlphaLoop method. This method does not take any parameters.

```
## Examples
```python
alphaloop_method = VakintEvaluationMethod.new_alphaloop_method()
```
```

new_matad_method

Kind: AssociatedFunction

```
def new_matad_method(cls, expand_masters: typing.Optional[builtins.bool] = None,
substitute_masters: typing.Optional[builtins.bool] = None, substitute_hpls:
typing.Optional[builtins.bool] = None, direct_numerical_substitution:
typing.Optional[builtins.bool] = None) -> VakintEvaluationMethod:
```

Create a new VakintEvaluationMethod instance representing the MATAD method.

```
## Examples
```python
matad_method = VakintEvaluationMethod.new_matad_method(
expand_masters=True,
substitute_masters=True,
substitute_hpls=True,
direct_numerical_substitution=True
)
```
```

Parameters

expand_masters : Optional[bool]
Whether to expand master integrals. Default is True.

substitute_masters : Optional[bool]
Whether to substitute master integrals. Default is True.

substitute_hpls : Optional[bool]
Whether to substitute harmonic polylogarithms. Default is True.

direct_numerical_substitution : Optional[bool]
Whether to perform direct numerical substitution. Default is True.

expand_masters

Kind: Parameter

expand_masters: typing.Optional[builtins.bool] = None

Default: None

Whether to expand master integrals. Default is True.

substitute_masters

Kind: Parameter

substitute_masters: typing.Optional[builtins.bool] = None

Default: None

Whether to substitute master integrals. Default is True.

substitute_hpls

Kind: Parameter

substitute_hpls: typing.Optional[builtins.bool] = None

Default: None

Whether to substitute harmonic polylogarithms. Default is True.

direct_numerical_substitution

Kind: Parameter

direct_numerical_substitution: typing.Optional[builtins.bool] = None

Default: None

Whether to perform direct numerical substitution. Default is True.

new_fmft_method

Kind: AssociatedFunction

```
def new_fmft_method(cls, expand_masters: typing.Optional[builtins.bool] = None,
substitute_masters: typing.Optional[builtins.bool] = None) ->
VakintEvaluationMethod:
```

Create a new VakintEvaluationMethod instance representing the FMFT method.

Examples

```
```python
fmft_method = VakintEvaluationMethod.new_fmft_method(
expand_masters=True,
substitute_masters=True
)
```
```

Parameters

expand_masters : Optional[bool]

Whether to expand master integrals. Default is True.

substitute_masters : Optional[bool]

Whether to substitute master integrals. Default is True.

expand_masters

Kind: Parameter

expand_masters: typing.Optional[builtins.bool] = None

Default: None

Whether to expand master integrals. Default is True.

substitute_masters

Kind: Parameter

substitute_masters: typing.Optional[builtins.bool] = None

Default: None

Whether to substitute master integrals. Default is True.

new_pysecdec_method

Kind: AssociatedFunction

```
def new_pysecdec_method(cls, quiet: typing.Optional[builtins.bool] = None,
relative_precision: typing.Optional[builtins.float] = None, min_n_evals:
```

```
typing.Optional[builtins.int] = None, max_n_evals: typing.Optional[builtins.int] =
None, reuse_existing_output: typing.Optional[builtins.str] = None, numerical_masses:
typing.Optional[typing.Mapping[builtins.str, builtins.float]] = None,
numerical_external_momenta: typing.Optional[typing.Mapping[builtins.int,
tuple[builtins.float, builtins.float, builtins.float, builtins.float]]] = None) ->
VakintEvaluationMethod:
```

Create a new VakintEvaluationMethod instance representing the numerical pySecDec method.

Examples

```
```python
pysecdec_method = VakintEvaluationMethod.new_pysecdec_method(
 quiet=True,
 relative_precision=1e-7,
 min_n_evals=10_000,
 max_n_evals=1_000_000_000_000,
 reuse_existing_output=None,
 numerical_masses={"muvsq": 1.0},
 numerical_external_momenta={1: (1.0, 0.0, 0.0, 0.0), 2: (0.0, 1.0, 0.0, 0.0)}
)
```
```

Note that for because pySecDec can only do numerical evaluations, the preset values of the masses and external momenta must be provided here.

Parameters

quiet : Optional[bool]

Whether to suppress output from pySecDec. Default is True.

relative_precision : Optional[float]

The relative precision to be achieved in the numerical integration. Default is 1e-7.

min_n_evals : Optional[int]

The minimum number of evaluations to be performed in the numerical integration. Default is 10,000.

max_n_evals : Optional[int]

The maximum number of evaluations to be performed in the numerical integration. Default is 1,000,000,000,000.

reuse_existing_output : Optional[str]

Path to existing pySecDec output to reuse. Default is None.

numerical_masses : Optional[Dict[str, float]]

A dictionary mapping mass parameter names to their numerical values. Default is an empty dictionary.

numerical_external_momenta : Optional[Dict[int, Tuple[float, float, float, float]]]

A dictionary mapping external momentum indices to their numerical 4-vector values. Default is an empty dictionary.

quiet

Kind: Parameter

quiet: typing.Optional[builtins.bool] = None

Default: None

Whether to suppress output from pySecDec. Default is True.

relative_precision

Kind: Parameter

relative_precision: typing.Optional[builtins.float] = None

Default: None

The relative precision to be achieved in the numerical integration. Default is 1e-7.

min_n_evals

Kind: Parameter

min_n_evals: typing.Optional[builtins.int] = None

Default: None

The minimum number of evaluations to be performed in the numerical integration. Default is 10,000.

max_n_evals

Kind: Parameter

max_n_evals: typing.Optional[builtins.int] = None

Default: None

The maximum number of evaluations to be performed in the numerical integration. Default is 1,000,000,000,000.

reuse_existing_output

Kind: Parameter

reuse_existing_output: typing.Optional[builtins.str] = None

Default: None

Path to existing pySecDec output to reuse. Default is None.

numerical_masses

Kind: Parameter

numerical_masses: typing.Optional[typing.Mapping[builtins.str, builtins.float]] = None

Default: None

A dictionary mapping mass parameter names to their numerical values. Default is an empty dictionary.

numerical_external_momenta

Kind: Parameter

numerical_external_momenta: typing.Optional[typing.Mapping[builtins.int, tuple[builtins.float, builtins.float, builtins.float, builtins.float]]] = None

Default: None

A dictionary mapping external momentum indices to their numerical 4-vector values. Default is an empty dictionary.

Exhaustive reference · Source

VakintExpression

A Vakint integral split into its numerator and normalized topology structure.

A Vakint integral split into its numerator and normalized topology structure.

Construct this wrapper from a Symbolica expression before applying Vakint operations.

class VakintExpression:

Requires features: symbolica_community_module, python_stubgen

Inspect the registered export

```
from symbolica.community.vakint import VakintExpression
help(VakintExpression)
```

`__str__`

Kind: Method

```
def __str__(self) -> builtins.str:
```

String representation of the VakintExpression.

`to_expression`

Kind: Method

```
def to_expression(self) -> Expression:
```

Convert the VakintExpression back to a Symbolica Expression.

Examples

```
```python
integral = VakintExpression(E('''
(
k(1,11)*k(1,11)
)*topo(
prop(1,edge(1,1),k(1),muvsq,1)
)''', default_namespace="vakint"))
str(integral)
(k(1,11)^2) x topo(prop(1,edge(1,1),k(1),muvsq,1))
```
```

`__new__`

Kind: AssociatedFunction

```
def __new__(cls, atom: typing.Any) -> VakintExpression:
```

Create a new VakintExpression from a Symbolica Expression which will separate numerator and topologies

Examples

```
```python
integral=E('''
(
k(1,11)*k(2,11)*k(1,22)*k(2,22)
+ p(1,11)*k(3,11)*k(3,22)*p(2,22)
+ p(1,11)*p(2,11)*(k(2,22)+k(1,22))*k(2,22)
)*topo(
prop(1,edge(1,2),k(1),muvsq,1)
* prop(2,edge(2,3),k(2),muvsq,1)
* prop(3,edge(3,1),k(3),muvsq,1)
)''')
```

```

* prop(4,edge(1,4),k(3)-k(1),muvsq,1)
* prop(5,edge(2,4),k(1)-k(2),muvsq,1)
* prop(6,edge(3,4),k(2)-k(3),muvsq,1)
)''' , default_namespace="vakint")
print(VakintExpression(integral))
#
((k(1,22)+k(2,22))*k(2,22)*p(1,11)*p(2,11)+k(1,11)*k(1,22)*k(2,11)*k(2,22)+k(3,11)*k(3,22)*p(1,11)*
x
topo(prop(1,edge(1,2),k(1),muvsq,1)*prop(2,edge(2,3),k(2),muvsq,1)*prop(3,edge(3,1),k(3),muvsq,1)*p
k(1)+k(3),muvsq,1)*prop(5,edge(2,4),k(1)-k(2),muvsq,1)*prop(6,edge(3,4),k(2)-
k(3),muvsq,1))
'''

```

Parameters

-----

atom : Expression

A Symbolica Expression containing a vakint integral, i.e. a sum of terms, each a product of a numerator and a ``vakint::topo(...)`` structure.

**atom**

**Kind:** Parameter

atom: typing.Any

A Symbolica Expression containing a vakint integral, i.e. a sum of terms, each a product of a numerator and a ``vakint::topo(...)`` structure.

Exhaustive reference · Source

### VakintNumericalResult

Container class storing the result of a numerical evaluation of a vakint expression as a Laurent series in the dimensional regularisation parameter epsilon.

Container class storing the result of a numerical evaluation of a vakint expression as a Laurent series in the dimensional regularisation parameter epsilon.

```
class VakintNumericalResult:
```

**Requires features:** symbolica\_community\_module, python\_stubgen

### Inspect the registered export

```
from symbolica.community.vakint import VakintNumericalResult
help(VakintNumericalResult)
```

**\_\_str\_\_**

**Kind:** Method

```
def __str__(self) -> builtins.str:
```

String representation of the numerical result.

## Examples

```
```python
result = VakintNumericalResult([
(-3, (0.0, -11440.53140354612)),
(-2, (0.0, 57169.95521898031)),
(-1, (0.0, -178748.9838377694)),
(0, (0.0, 321554.1122184795)),
])
```
```

```
str(result)
 ϵ^{-3} : (0+-11440.5314035461i)
 ϵ^{-2} : (0+57169.9552189803i)
 ϵ^{-1} : (0+-178748.983837769i)
 ϵ^0 : (0+321554.112218480i)
```

### **to\_list**

**Kind:** Method

```
def to_list(self) -> builtins.list[tuple[builtins.int, tuple[builtins.float,
builtins.float]]]:
```

Convert the numerical result to a native Python list of (epsilon exponent, (real, imag)) tuples.

## Examples

```
```python
result = VakintNumericalResult([
(-3, (0.0, -11440.53140354612)),
(-2, (0.0, 57169.95521898031)),
(-1, (0.0, -178748.9838377694)),
(0, (0.0, 321554.1122184795)),
])

result.to_list()
# [(-3, (0.0, -11440.53140354612)), (-2, (0.0, 57169.95521898031)), (-1, (0.0,
-178748.9838377694)), (0, (0.0, 321554.1122184795))]
```
```

### **\_\_new\_\_**

**Kind:** AssociatedFunction

```
def __new__(cls, values: typing.Sequence[tuple[builtins.int, tuple[builtins.float,
builtins.float]]]) -> VakintNumericalResult:
```

Create a new instance of VakintNumericalResult from a list of (epsilon exponent, (real, imag)) tuples.

## Examples

```
```python
VakintNumericalResult([
(-3, (0.0, -11440.53140354612)),
(-2, (0.0, 57169.95521898031)),
(-1, (0.0, -178748.9838377694)),
(-0, (0.0, 321554.1122184795)),
])
```
```

### **Parameters**

-----

values : List[Tuple[int, Tuple[float, float]]]

A list of tuples, each containing an integer exponent of epsilon and a tuple of two floats representing the real and imaginary parts of the coefficient.

## **values**

**Kind:** Parameter

values: typing.Sequence[tuple[builtins.int, tuple[builtins.float, builtins.float]]]

A list of tuples, each containing an integer exponent of epsilon and a tuple of two floats representing the real and imaginary parts of the coefficient.

## **compare\_to**

**Kind:** Method

```
def compare_to(self, other: VakintNumericalResult, relative_threshold:
builtins.float, error: typing.Optional[VakintNumericalResult] = None, max_pull:
typing.Optional[builtins.float] = None) -> tuple[builtins.bool, builtins.str]:
```

Compare this numerical result to another, returning a tuple of (bool, str) where the bool indicates whether the results match within the specified thresholds, and the str provides details of the comparison.

## Examples

```
```python
result1 = VakintNumericalResult([
(-3, (0.0, -11440.53140354612)),
])
result2 = VakintNumericalResult([
(-3, (0.0, -11440.53140354612)),
(-2, (0.0, 2.0)),
])
result1.compare_to(result2, relative_threshold=1e-5)
# (False, 'imaginary part of  $\epsilon^{-2}$  coefficient does not match within rel. error
required: 0 != 2.0000000000000000 (rel. error = 2.0000000000000000)')
```
```

## Parameters

-----

other : VakintNumericalResult

The other numerical result to compare to.

relative\_threshold : float

The relative threshold for comparison.

error : Optional[VakintNumericalResult]

An optional numerical result representing the error in the evaluation.

max\_pull : Optional[float]

The maximum pull for comparison. Default is 3.0.

## **other**

**Kind:** Parameter

other: VakintNumericalResult

The other numerical result to compare to.

## **relative\_threshold**

**Kind:** Parameter

relative\_threshold: builtins.float

The relative threshold for comparison.

**error****Kind:** Parameter`error: typing.Optional[VakintNumericalResult] = None`**Default:** None

An optional numerical result representing the error in the evaluation.

**max\_pull****Kind:** Parameter`max_pull: typing.Optional[builtins.float] = None`**Default:** None

The maximum pull for comparison. Default is 3.0.

[Exhaustive reference](#) · [Source](#)

## Generated topology and dependency reference

These tables come from Vakint's runtime topology generator and minimum-version constants.

### External dependencies

| Dependency | Minimum | Source constant          |
|------------|---------|--------------------------|
| FORM       | 4.2.1   | MINIMAL_FORM_VERSION     |
| pySecDec   | 1.6.4   | MINIMAL_PYSECDEC_VERSION |

### Supported topology patterns

| Name                 | Loops | Propagator slots |
|----------------------|-------|------------------|
| I1L                  | 1     | 1                |
| I2L                  | 2     | 3                |
| I2L_pinch_3          | 2     | 3                |
| I3L                  | 3     | 6                |
| I3L_pinch_6          | 3     | 6                |
| I3L_pinch_1_6        | 3     | 6                |
| I3L_pinch_3_6        | 3     | 6                |
| I3L_pinch_1_3_6      | 3     | 6                |
| I4L_H                | 4     | 9                |
| I4L_X                | 4     | 9                |
| I4L_BMW              | 4     | 8                |
| I4L_FG               | 4     | 8                |
| I4L_FG_pinch_2       | 4     | 8                |
| I4L_FG_pinch_3       | 4     | 8                |
| I4L_FG_pinch_4       | 4     | 8                |
| I4L_FG_pinch_7       | 4     | 8                |
| I4L_FG_pinch_8       | 4     | 8                |
| I4L_FG_pinch_1_8     | 4     | 8                |
| I4L_FG_pinch_2_4     | 4     | 8                |
| I4L_FG_pinch_4_7     | 4     | 8                |
| I4L_FG_pinch_4_8     | 4     | 8                |
| I4L_FG_pinch_7_8     | 4     | 8                |
| I4L_FG_pinch_2_4_7   | 4     | 8                |
| I4L_FG_pinch_3_4_8   | 4     | 8                |
| I4L_FG_pinch_4_7_8   | 4     | 8                |
| I4L_FG_pinch_6_7_8   | 4     | 8                |
| I4L_FG_pinch_4_6_7_8 | 4     | 8                |

## Snapshot and version metadata

This manual was rendered from one immutable workspace identity.

- **Channel:** latest
- **Route:** /products/vakint/latest/
- **Git commit:** 188ef8559ff727ab7b3b78827cca0a502880f1ce

## Workspace component versions

- gammaloop/gammalooprs — 0.3.4 (features: none (Cargo defaults disabled))
- gammaloop/gammaloop-api — 0.1.0 (features: cli (Cargo defaults disabled; complete explicit matrix))
- gammaloop/gammaloop — 0.3.4 (features: python\_api, python\_abi, python\_stubgen (Cargo defaults disabled; complete explicit matrix))
- linnet/linnet — 0.17.0 (features: nodestore-vec, serde, bincode, drawing, symbolica (Cargo defaults disabled; complete explicit matrix))
- linnet/linnet-py — 0.1.0 (features: extension-module, abi3-py310 (Cargo defaults disabled; complete explicit matrix))
- spenso/spenso — 0.6.0 (features: shadowing (Cargo defaults disabled; complete explicit matrix))
- spenso/spenso-macros — 0.3.2 (features: shadowing (Cargo defaults disabled; complete explicit matrix))
- spenso/spenso-hep-lib — 0.1.7 (features: none (Cargo defaults disabled))
- spenso/spynso3 — 0.1.2 (features: python\_stubgen (Cargo defaults disabled; complete explicit matrix))
- idenso/idenso — 0.3.0 (features: bincode, reference-cases (Cargo defaults disabled; complete explicit matrix))
- idenso/idenso — 0.3.0 (features: python, python\_stubgen (Cargo defaults disabled; complete explicit matrix))
- vakint/vakint — 0.1.2 (features: none (Cargo defaults disabled))
- vakint/vakint — 0.1.2 (features: symbolica\_community\_module, python\_stubgen (Cargo defaults disabled; complete explicit matrix))